

BeamFormer: Transformer-based Beam Management for 6G Networks

Shunqiang Feng
shunqiang@virginia.edu
University of Virginia

Swastik Kanjilal
kanjis2@rpi.edu
RPI

Kun Qian
kunqian@virginia.edu
University of Virginia

Ish Kumar Jain
jaini@rpi.edu
RPI

Abstract

The forthcoming 6G wireless network is anticipated to augment base stations with much denser antenna arrays, achieving ultra-narrow beamforming and orders of magnitude higher capacity than 5G. However, the resulting beam management for such antenna arrays imposes prohibitive overhead, as the number of beam measurements needed for full directional coverage increases sharply. Inspired by recent advances in generative models, we propose BEAMFORMER, a transformer-based framework that generates full beam spectra from a few reference beam measurements. The success of applying transformers to accurate, fast, and robust beam management stems from joint architectural innovations, including beam pattern encoding, latent beam processing with linear complexity, and universal reference beam optimization. We synthesize a large channel dataset to train BEAMFORMER, and evaluate it using both the simulation dataset across multiple FR2/FR3 frequencies and the real dataset from 28 GHz software-defined radio testbeds. Compared to existing approaches, BEAMFORMER achieves better beam alignment with an average gain of 6.7 dB. Moreover, BEAMFORMER achieves a 1.8 ms delay to generate a beam spectrum with 1,600 beams, empowering real-time beam management for future 6G. The source code and dataset are publicly accessible at GitHub¹.

CCS Concepts

• **Networks** → **Mobile networks**; • **Computing methodologies** → **Neural networks**; • **Hardware** → **Wireless devices**.

ACM Reference Format:

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1 Introduction

Emerging high-bandwidth applications, such as hologram telepresence, collaborative robots, and human augmentation, are accelerating the evolution of wireless network infrastructure. To meet the resulting capacity demands, the transition from 5G to 6G is marked by two critical technological shifts. First, 6G plans to utilize FR3 bands (7125 – 24250 MHz) [1], which offer more spectrum resources than the FR1 bands (410 – 7125 MHz) and provide wider coverage

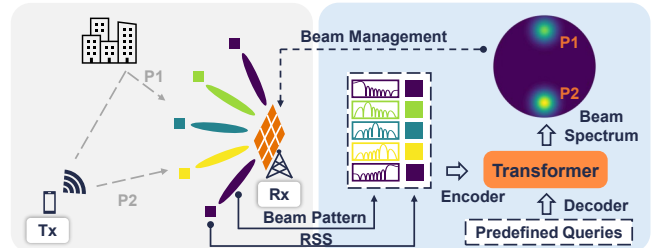


Figure 1: Applications of BEAMFORMER in beam management for 6G networks

than the FR2 bands (24250 – 71000 MHz) used in current 5G. Second, 6G will compact more antenna elements proportionally on the same base station, known as Giga MIMO (gMIMO) [42], to create ultra-narrow beams that simultaneously boost per-link capacity and multi-link spatial reuse. Denser antenna arrays, combined with ample FR3 spectrum resources, enable orders of magnitude higher capacity than 5G.

Despite the high capacity, the massive number of antennas in gMIMO will dramatically escalate the hardware and operational costs of 6G. To manage the hardware complexity, 6G base stations must group antenna elements into phased arrays, each connected to a single RF chain, and perform analog beamforming within each array. To steer the beam to a user equipment, the base station must sweep across all candidate beams and measure the received signal strength (RSS). Unfortunately, such beam management overhead grows substantially with the number of antennas and is exacerbated in mobile scenarios [18, 51], e.g., vehicular communications, where dynamic channels require more frequent beam realignment. While hierarchical beam sweeping has been adopted in existing beam management protocols [16, 35] to reduce overhead, the multiple rounds of beam search still introduce non-negligible latencies [20]. In addition, the presence of incumbent services in the FR3 band, e.g., satellites and radars, further complicates the beam management process of 6G. Specifically, interference constraints limit the set of usable beams, resulting in insufficient channel measurements and suboptimal link performance.

In contrast to hierarchical beam sweeping, some works [38, 49] propose to recover the channel state information (CSI) per antenna from RSS measurements of specially designed beams. Then, the optimal beam configuration can be derived, enabling single-round beam alignment. These approaches are mainly based on compressive sensing [26, 27, 43, 47, 61], which utilize the inherent sparsity of wireless channels to reduce the number of beam measurements. Despite this, their beam measurement overhead is still significant. For example, 2ACE [49] requires 384 beam measurements to estimate the CSI for a 256-element transmitter and a single-element receiver. It further incurs second-level delays in reconstructing CSI, which is incompatible with the stringent latency requirement in

¹<https://github.com/Shunqiang-Feng/BeamFormer.git>



practical beam management processes.

Regarding the objective of beam alignment, reconstructing the complete CSI is not only time-consuming but also excessive. Instead, it suffices to identify the directions of dominant propagation paths and the signal strengths of the beams aligned with them. To this end, we aim to reconstruct *beam spectra*, which depict the RSS of any beam. Since pencil beams are commonly used and feature highly directional main lobes, beam spectrum reconstruction is analogous to image inpainting [15, 41] in computer vision, which aims to recover a complete image from a subset of pixels sampled along some directions. Inspired by this connection, we propose BEAMFORMER, a transformer-based beam management model, which generates the RSS of unknown query beams given RSS measurements of a few reference beams.

Based on the attention mechanism, generative transformers [50] have demonstrated superior performance in generating images [13] and texts [50]. BEAMFORMER further extends the transformers to beam spectrum generation. Fig. 1 shows the application of BEAMFORMER for beam management in 6G networks. Specifically, the base station measures the RSS of a small set of reference beams, which, together with the beam configurations, are input into a BEAMFORMER as the “prompt.” Then, the base station queries BEAMFORMER about the RSS of interesting beams, reconstructs the beam spectrum, and identifies the optimal beam aligned with the dominant propagation path. Note that BEAMFORMER is distinct from existing RF neural representations, e.g., NeRF [59], which are only applicable to a single scene. Instead, BEAMFORMER captures the inherent relations between different beams and is naturally generalizable to different scenarios.

However, applying transformers to beam data is not intuitive, and three key challenges must be overcome.

(i) *How to encode beam information for a transformer?* Both reference beam RSS and configurations are important information for beam spectrum generation. Given the structure of a standard transformer [13, 50], it is natural to use beam RSS as data embeddings and beam configurations as positional encodings. While common pencil beams have dominant main lobes, whose directions can be used as positional encodings, such a simple scheme loses key information about beam configurations. Specifically, unlike cameras, where each pixel corresponds to a perfect ray from a specific direction, the beam patterns of antenna arrays often have strong sidelobes that can also capture prominent signals. Besides, base stations may use multi-armed beams with multiple main lobes [20]. Beam patterns also depend on the array geometry, which varies across different hardware. To overcome these limitations, we introduce a beam pattern encoding scheme that produces informative and generalized positional encodings for transformers.

(ii) *How to accelerate the generation process for practical beam management in 6G networks?* Standard transformer models are based on the attention mechanism, which has quadratic complexity with respect to the number of input tokens [52]. It limits the number of query beams per beam management cycle, thereby degrading the resolution of the generated beam spectrum. To reduce model complexity, we apply a transformer variant with latent space processing [24] to efficiently encode the reference beam measurements and decode the query beams. Using a fixed-size latent space aligns with the inherent sparsity of wireless channels characterized by

a limited number of multipath components. By processing in the latent space, the model complexity scales linearly with the number of query beams, enabling efficient beam spectrum generation.

(iii) *How to mitigate the generation inconsistency of a transformer?* Unrealistic generation is one of the most critical problems of transformers. For example, a transformer may inpaint phantom objects or incorrect colors in images [53]. Such inconsistent generation is mainly due to insufficient or biased prompts. Fortunately, unlike image inpainting with uncontrollable reference pixels, beam spectrum generation allows customization of reference beams. To leverage the beam reconfigurability of base stations, we propose a learnable beam generation module that identifies the universally optimal reference beams.

We implement BEAMFORMER and evaluate it using the simulation data generated by RF ray tracing at common FR2/FR3 frequency bands, i.e., 14 GHz, 28 GHz, and 60 GHz. We also test BEAMFORMER in realistic outdoor and indoor scenarios using two 28 GHz software-defined radios (SDR) featuring phased arrays. BEAMFORMER achieves an average gain of 6.7 dB, compared to existing beam management approaches. Moreover, it achieves a delay of only 1.8 ms to generate a beam spectrum with 1,600 beams, potentially enabling real-time beam management for 6G.

In summary, our main contributions are:

(i) To the best of our knowledge, this work is the first to investigate the generative capability of transformers for beam measurement estimation. The resulting BEAMFORMER overcomes the prohibitive overhead of beam alignment imposed by gMIMO and spectrum coexistence in 6G.

(ii) We develop core components, including beam pattern encoding, latent beam processing, and universal reference beam optimization, to adapt transformers to the new data modality of beam measurements.

(iii) Through comprehensive simulations and experiments, we show that BEAMFORMER achieves high accuracy, low latency, and versatility across various configurations. BEAMFORMER has the potential to satisfy the stringent beam management requirements of future 6G networks. Note that BEAMFORMER is designed solely for beam management at the PHY layer, and full-stack processing is beyond its scope.

2 Preliminary

2.1 Phased Array Beamforming

A phased array consists of multiple antenna elements, each capable of adjusting the amplitude and phase of the signal it radiates. The resulting superimposed signal exhibits unique power distribution across directions, known as a beam pattern. As shown in Fig. 2, for a direction determined by an azimuth angle $\phi \in [0, 2\pi]$ and an elevation angle $\theta \in [0, \pi/2]$, the gain of the phased array is:

$$g(\phi, \theta) = \vec{w}^T \vec{s}(\phi, \theta), \quad (1)$$

where $\vec{w} = [\alpha_1 e^{j\theta_1}, \alpha_2 e^{j\theta_2}, \dots, \alpha_N e^{j\theta_N}]^T$ is the weight vector of amplitude and phase adjustments of all N antennas, and $\vec{s}(\phi, \theta) = [e^{j\frac{2\pi}{\lambda} \vec{p}_1^T \hat{v}}, e^{j\frac{2\pi}{\lambda} \vec{p}_2^T \hat{v}}, \dots, e^{j\frac{2\pi}{\lambda} \vec{p}_N^T \hat{v}}]^T$ is the steering vector. In the steering vector $\vec{s}$, $\vec{p}_n = [x_n, y_n, z_n]^T$ is the location of the n -th antenna in the phased array, and $\hat{v} = [\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta]^T$ is the unit direction vector.

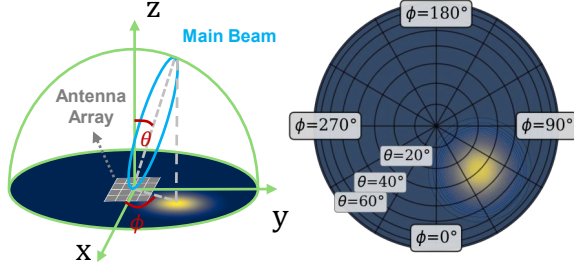


Figure 2: Illustration of a beam pattern with the main beam towards $(\phi, \theta) = (45^\circ, 30^\circ)$.

Due to the multipath effect, the received signal is the superimposition of the signals from several directions:

$$r = \sum_{k=1}^K \alpha_k g(\phi_k, \theta_k), \quad (2)$$

where α_k is the attenuation and phase shift along the k -th propagation path, and (ϕ_k, θ_k) is the angle of departure (AoD). The RSS $R = |r|^2$. Note that when the channel is fixed, the RSS varies with the beam weight vector $\vec{w}$.

2.2 Beam Sweeping

To maximize the link throughput, both the transmitter and the receiver need to find the optimal beam weight vectors to maximize the RSS. Intuitively, it is achieved when the main lobes of both sides align with each other. Beam sweeping is commonly employed for beam alignment, wherein the transmitter and receiver sequentially measure the RSS of a set of predefined beam weight vectors and identify the beam pair that yields the maximum RSS.

For example, in the IEEE 802.11ad/ay (WiGig) [35], beam sweeping is performed during the Beacon Header Interval (BHI) of each Beacon Interval (BI), where a set of Sector Sweep (SSW) frames with different beam weights is sent from one side, while the other side retains the quasi-omnidirectional pattern. WiGig employs a hierarchical beam sweeping scheme by conducting beam refinement during the Data Transmission Interval (DTI) of each BI, thereby reducing the beam searching overhead. On the other hand, 5G mmWave New Radio adopts a conceptually similar hierarchical beam management framework. Specifically, coarse beam sweeping is conducted through Synchronization Signal/PBCH block (SSB) bursts during initial access, while beam refinement and tracking are performed using reference signals (RS) with beamforming [30]. Beam sweeping is fundamental to mmWave communication and is anticipated to be equally important for 6G in the FR3 band, as it circumvents the need for time-consuming CSI estimation, thereby achieving a practical trade-off between channel measurement accuracy and efficiency.

Beam sweeping yields samples of the beam spectrum, *i.e.*, the distribution of RSS across beam weight vectors:

$$\vec{R} = [R(\vec{w}_1), R(\vec{w}_2), \dots, R(\vec{w}_M)]. \quad (3)$$

Since common beam patterns only have a dominant main lobe, *i.e.*, pencil beams, the RSS measurements of the beams can be assigned to the corresponding main lobe directions, forming an angular spectrum, as shown in Fig. 1. Intuitively, denser sampling of the beam spectrum through more beam measurements leads to better beam

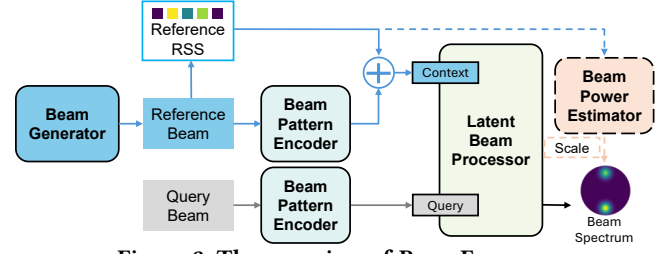


Figure 3: The overview of BEAMFORMER.

alignment. However, as antenna array sizes grow significantly in 6G, the latency associated with beam sweeping becomes prohibitive.

2.3 Transformer

Transformers were originally proposed for natural language processing [50] and have since demonstrated strong generative capability across diverse modalities, including images [13] and audio [17]. The key innovation is the attention mechanism, which captures long-range dependencies among input data units, *i.e.*, tokens, by computing pairwise interactions. Each token is represented by a feature vector that comprises the encoding of the token's position in the input token sequence and the embedding of the token's data. A transformer model is composed of stacked attention layers, each computing attention scores between query and key tokens and updating the query tokens to reflect their relative importance, *i.e.*,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \quad (4)$$

where Q , K , and V denote the query, key, and value matrices, respectively, and d is the dimension of the key vectors. In standard transformer self-attention, Q , K , and V representations are derived from the same input token sequence. Therefore, the computational complexity scales quadratically with the number of input tokens.

3 System Overview

BEAMFORMER addresses the critical beam management overhead associated with the gMIMO in the upcoming 6G. Compared to beam sweeping, BEAMFORMER generates the complete beam spectrum from the RSS measurements of a small group of reference beams. It is similar to the “image inpainting” task [15, 41] in computer vision, where a group of pixel patches is used to recover a complete image. Note that each pixel in an image represents the light intensity from a corresponding direction, similar to the RSS of a beam. Inspired by this connection, we introduce the state-of-the-art generative models, *i.e.*, transformer [13, 50], to generate beam spectrum.

BEAMFORMER innovates in three key modules to adapt the standard transformer to beam spectrum data, as highlighted in Fig. 3. First, BEAMFORMER uses beam patterns as positional encodings to accommodate arbitrary beam configurations, *e.g.*, multiple main lobes and prominent side lobes, as well as various antenna array configurations, *e.g.*, different antenna numbers, geometries, and weight quantization. The *Beam Pattern Encoder* module converts the discretized beam patterns into positional encodings for the transformer (§ 4.1). Second, BEAMFORMER adopts a transformer variant that encodes input tokens into a fixed-size latent space before self-attention. Therefore, the module, named as *Latent Beam Processor*, achieves linear complexity and real-time beam spectrum genera-

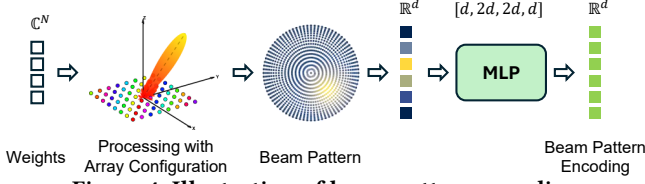


Figure 4: Illustration of beam pattern encoding

tion (§ 4.2). Third, BEAMFORMER optimizes the reference beams to mitigate transformer generation inconsistency. The *Beam Generator* is a learnable module jointly trained with the main transformer model to obtain optimal reference beams and maximize generation quality (§ 4.3). Additionally, the transformer typically processes normalized inputs. Therefore, BEAMFORMER uses a *Beam Power Estimator* module to recover the absolute RSS (§ 4.4).

BEAMFORMER can be integrated into the existing PHY layer beam management process of wireless networks. First, as in the default beam sweeping process, the base station measures the RSS of the reference beams generated by the Beam Generator. The beam patterns are fed into the Beam Pattern Encoder to generate positional encodings, and the RSS are normalized and projected through a linear layer to obtain the embeddings. To obtain the RSS of query beams, the corresponding beam patterns are encoded to create query tokens. Then, the Latent Beam Processor processes the tokens and outputs the RSS for all query beams, which the Beam Power Estimator denormalizes to form the final beam spectrum. In line with typical cellular infrastructure, BEAMFORMER focuses on single-sided beam management at base stations, since user equipment tends to have small antenna arrays. A base station usually has multiple antenna arrays, each serving a different user equipment. Therefore, the base station can create a BEAMFORMER instance for each antenna array and the user equipment it serves.

4 Design of BEAMFORMER

4.1 Beam Pattern Encoding

BEAMFORMER converts beam RSS measurements into embeddings for the transformer. Each embedding requires a positional encoding to describe its position within the input. The sequential orders of the input tokens along multiple dimensions are typically used for positional encoding [31, 60]. Therefore, an intuitive encoding scheme is to use the direction of the main lobe, *i.e.*, azimuth and elevation, of the beam patterns. However, this scheme omits critical factors that affect beam RSS measurements. Specifically, RSS depends not only on the direction of the main lobe, but also on the gain and width of the main lobe, as well as the prominent side lobes, as in Eq. 2. These factors are essentially determined by the array geometry, *e.g.*, number of antennas, antenna position, polarization, and radiation pattern, which may vary across hardware. In addition, more sophisticated beam patterns with multiple main lobes can be used in some beam management schemes [20], making the simple direction encoding scheme infeasible.

Compared with the main lobe direction, the beam pattern captures all factors that affect RSS and is agnostic to antenna array geometries. Therefore, we design a Beam Pattern Encoder that generates positional encodings from beam patterns, thereby avoiding the need for model modifications or retraining across different

antenna array hardware. As defined in Eq. 1, a beam pattern characterizes the gain of an antenna array across different directions when a beam weight vector is applied. We discretize the beam pattern and normalize its amplitude to facilitate neural network processing. While the actual beam pattern also depends on the radiation pattern of each antenna, we only use an isotropic radiation pattern for simplicity of data synthesis.

Fig. 4 illustrates the workflow of beam pattern encoding. Given the beam weight vector $\mathbf{w} \in \mathbb{C}^N$ of an N -element array, BEAMFORMER computes the corresponding beam pattern at the sample directions. Specifically, we uniformly sample the continuous beam pattern at 64 azimuth directions and 16 elevation directions, yielding an angular resolution of $5.5^\circ \times 5.5^\circ$. This resolution is chosen to be finer than the 3-dB beamwidth of a 16×16 URA (*i.e.*, 6.4°), allowing fine-grained features of the beam pattern to be captured without incurring unnecessary computational overhead. The 1,024 beam pattern samples are then flattened into a vector and passed through the Beam Pattern Encoder, *i.e.*, a multi-layer perceptron (MLP), to obtain the positional encoding. The Beam Pattern Encoder consists of four fully connected layers with dimensions d , $2d$, $2d$, and d . If the array becomes even larger, denser beam pattern samples can be used to achieve a finer resolution. By keeping the feature dimensions low, the overall computational cost can still remain manageable.

4.2 Latent Beam Processing

BEAMFORMER was initially designed using the standard transformer model. However, these models suffer from severe scalability limitations. In particular, the transformer forces both reference and query beams to pass through every self-attention layer, which incurs quadratic complexity and renders it computationally infeasible to output thousands of beam measurements for beam spectra generation within the stringent time limitation of beam management.

Recent transformer variants introduce latent space to decouple the processing of context and query, thereby reducing the computational overhead of the transformer [24, 25]. Specifically, the context tokens are encoded into a fixed-size latent space and processed, while the queries only attend to the final latent representations. Therefore, the computational complexity is reduced to linear. The rationale of latent space processing lies in the substantial redundancy of raw data inputs, *e.g.*, text, image, and audio, whose key information can thus be condensed into fewer tokens in the latent space. It also aligns with the intrinsic sparsity of the high-frequency channel in FR2/FR3 bands. Specifically, a FR2/FR3 channel typically has only a few dominant paths, making reference beam measurements inherently redundant. Therefore, we replace the standard transformer model with the latent space processing variant in BEAMFORMER to accelerate beam spectrum generation while still preserving the fidelity of the reconstructed sparse beam spectra.

Fig. 5 illustrates the workflow of latent beam processing of BEAMFORMER. Compared to the standard transformer, BEAMFORMER uses an array of learnable latent tokens as a compact set of bases for sparse channel modeling, which can be optimized to extract key channel structure from the context tokens, *i.e.*, reference beam patterns and RSS measurements, via a cross attention layer. Then, the latent tokens are further processed via multiple self-attention layers

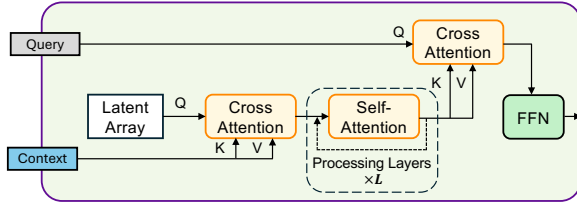


Figure 5: Illustration of latent beam processing

to progressively enhance the key features of the channel structure, e.g., separating dominant propagation paths from different directions and suppressing spectral noises. After processing in the latent space, the latent features are projected to the query tokens, *i.e.*, the query beam pattern encodings, via a second cross attention layer. Note that since the processing of all query tokens can be parallelized, the beam spectrum generation with the Latent Beam Processor can achieve sublinear complexity in practice. Beyond its computational efficiency, the latent beam processing workflow also reflects the asymmetry between the context and query, *i.e.*, the reference beams embed rich angular information of the multipath channel and thus require extensive processing, while the query beams used to construct the beam spectrum mainly only contain the main lobe direction information and can be processed in a lightweight manner.

The query outputs are eventually fed into a feed-forward network (FFN) to generate predicted beam RSS. The FFN is a two-layer MLP with an expansion ratio of 4 and a GEGLU activation layer. By default, the Latent Beam Processor predicts the RSS of 1,600 pencil beams pointing in different directions, forming a discrete beam spectrum with 20 elevation angles and 80 azimuth angles. Note that the 20×80 angular resolution of the beam spectrum is higher than that of the beam pattern, *i.e.*, 16×64 , thereby providing sufficiently fine directional granularity for a 16×16 URA.

To ensure stable training, the predicted beam spectrum is further passed through a Min-Max normalization layer, and the corresponding ground-truth spectra are normalized in a similar manner for supervision. The normalization process constrains the output spectrum to represent only the relative amplitude distribution, thereby mitigating the impact of orders-of-magnitude variations of practical signal attenuation. The absolute RSS, which is required for link quality assessment, is separately recovered by the Beam Power Estimator introduced in Section 4.4.

4.3 Universal Beam Optimization

BEAMFORMER leverages the generative capability of the transformer to predict the beam spectrum, whose correctness depends on the model capacity and the context informativeness [8]. However, the stringent time requirement of beam management significantly constrains both the model size and the maximum number of reference beam measurements. For example, BEAMFORMER typically uses 32 reference beams to reconstruct a beam spectrum with 1,600 query beams, resulting in a filling ratio of 2%. For reference, image inpainting with a low filling ratio of 5% already experiences poor generation quality, especially when the provided patches lack key context information [21]. Similarly, inappropriate reference beams may fail to reveal the dominant multipath structure, resulting in ambiguous and inconsistent beam spectrum generation. Fortunately,

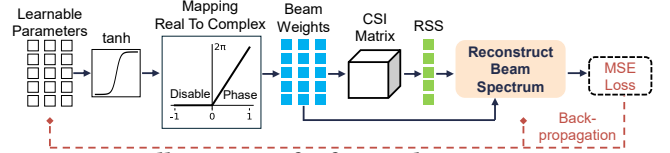


Figure 6: Illustration of reference beam optimization

unlike the image inpainting task, where the image patches fed to the transformer cannot be controlled, BEAMFORMER can customize the reference beams by controlling the corresponding beam weight vectors to improve the quality of the beam spectra generated. To find the universal optimal set of reference beams, we introduce the Beam Generator, a learnable module that jointly optimizes the reference beam patterns with other modules.

Fig. 6 illustrates the workflow of the Beam Generator. Specifically, it parameterizes the reference beams through an $M \times N$ learnable matrix, where M denotes the number of reference beams and N is the number of antennas. Each parameter is mapped to an antenna weight of a beam. However, due to the hardware limitation, the achievable beam weights of a phased array are limited. For example, a phased array typically does not support amplitude adjustment and only has quantized phase shifts. As a result, directly optimizing complex-valued beam weights may result in infeasible beams. Therefore, we need to establish a compact mapping between constrained beam weights and continuous learnable parameters to achieve seamless optimization.

To address the amplitude constraint, we introduce two differentiable layers to map the parameters to binary amplitudes, corresponding to activation and deactivation of antenna elements. First, a Tanh layer is used to map unconstrained real-valued parameters to values between -1 and 1. Then, a ReLU layer is used to emulate the activation of the antenna elements: if the Tanh layer output is smaller than 0, the corresponding antenna is deactivated; otherwise, if the Tanh layer output is between 0 and 1, the corresponding antenna is activated, and the phase shift varies between 0 and 2π accordingly. To address the phase quantization, we further incorporate the Straight-Through Estimator (STE) into the training process, which passes gradients through non-differentiable components by replacing zero derivatives with smooth surrogates during the backward propagation [6].

The Beam Generator is interfaced and jointly trained with the other modules of BEAMFORMER. As shown in Fig. 6, BEAMFORMER uses the beam weights and the CSI matrix available to simulated data samples to calculate the beam pattern and RSS measurements, which are passed through the Beam Pattern Encoder and the Latent Beam Processor. During the joint training, the Beam Generator learns reference beam weight vectors that minimize the beam spectrum reconstruction error. On the other hand, the Beam Pattern Encoder and the Latent Beam Processor are updated conditioned on the optimized reference beams. The joint training yields a universally optimal reference beamforming strategy that maximizes channel information given a limited number of beams, thereby enhancing beam spectrum generation.

In addition, the Beam Generator acts as an abstraction layer of phased array hardware details for other modules, thereby enhancing the scalability and deployability of BEAMFORMER. For example, migrating BEAMFORMER to new phased array hardware only re-

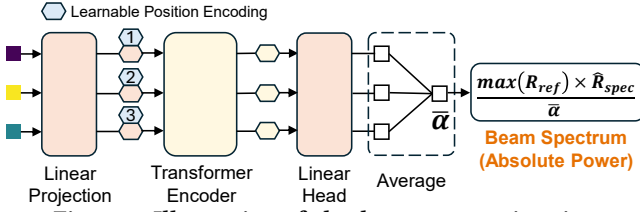


Figure 7: Illustration of absolute power estimation

quires retraining the Beam Generator to find the reference beam weight vectors for the new array geometry and fine-tuning other modules with the updated reference beam patterns.

4.4 Absolute Power Estimation

Beam management requires absolute RSS measurements for downstream link control tasks, *e.g.*, SNR estimation, rate adaptation, and power allocation. However, to ensure stable and convergent model training, the latent beam processing only generates normalized beam spectra. To recover the absolute RSS measurements, a straightforward solution is to reuse the normalization factor of the input, *i.e.*, the reference beam measurements. However, it requires the maximum reference beam measurement to be close to the global maximum, which is achieved only with a pencil beam aligned with the most dominant propagation path. Unfortunately, due to the limited number of reference beams, such a requirement is usually difficult to meet.

Instead of using only the maximum reference beam measurement, we leverage all reference beams and extract information about the absolute signal power from their RSS variations. Specifically, Fig. 7 illustrates the Beam Power Estimator, which incorporates a lightweight transformer encoder to predict the ratio of the maximum reference beam measurement to the global maximum of the beam spectrum. First, the RSS values are linearly projected into a 512-dimensional feature space and combined with learnable positional encodings. Then, the self-attention layers of the transformer encoder aggregate features across reference beam measurements. Finally, a linear head outputs multiple predictions of the ratio, which are averaged to obtain the final ratio $\bar{\alpha}$, thereby improving robustness across beam data in different scenarios. With the ratio estimate $\bar{\alpha}$, the maximum reference beam RSS $\max(R_{ref})$, and the normalized beam spectrum $\hat{R}_{spec}$ from the Latent Beam Processor, the absolute beam spectrum can be recovered, as illustrated in Fig. 7. Note that while the Beam Power Estimator uses a transformer encoder with quadratic complexity, the actual overhead is negligible due to the small number of reference beams.

5 Implementation

This section introduces the implementation details of BEAMFORMER, including the datasets and training procedures.

5.1 Simulation Dataset Generation

Based on the transformer, BEAMFORMER requires large training datasets with ground-truth beam spectra, which are impractical to collect with real hardware. Therefore, we resort to RF simulators [14, 23] to generate sufficient data samples.

Simulation setup. We use Wireless InSite [14], a commercial 3D ray-tracing software, to generate the simulation datasets. Multi-

Scenario	Indoor	Outdoor
Type	Home Office Classroom	Campus Urban Rural
Center Frequency	28 GHz	14/28/60 GHz
RT Condition	6r,1d,1t	6r,1d,0t
Array	16×16 URA, spacing = 0.5 λ	

Table 1: Summary of the simulation dataset

ple simulation cases are considered, and their configurations are summarized in Tab. 1.

The ray-tracing simulator requires mesh models of the scenarios and the transceivers placed within them. For indoor scenarios, we use a few open-source mesh models of varying types, *e.g.*, home, office, and classroom, each furnished with a range of objects to make the scenarios realistic. The transceivers are randomly placed at different locations and heights in open space. They are also rotated to face various directions. For outdoor scenarios, we extract building geometries, terrain, and road networks from OpenStreetMap to construct city mesh models, including campus, urban, and rural regions. The material of each building is randomly selected from a few common material types, including metal, glass, concrete, and brick. To emulate outdoor cellular networks, we place the base stations randomly within a predefined height range, *i.e.*, 1 – 25 m. In contrast, the user equipment is placed on vehicles and pedestrians on the ground. The vehicular and pedestrian traffic patterns are generated with the Simulator of Urban Mobility (SUMO) [28].

We generate simulation datasets for common FR2/FR3 frequency bands, *i.e.*, 14 GHz (6G FR3), 28 GHz (5G FR2), and 60 GHz (WiGig). For each frequency band, we simulate the CSI of 128 subcarriers to capture comprehensive channel characteristics. RSS is calculated by averaging the signal powers of all subcarriers and is thus less prone to the fast fading effects of multipath channels.

In both indoor and outdoor scenarios, we set the maximum number of reflections and diffractions to 6 and 1, respectively. In addition, we turned on diffuse scattering in the simulation and models near-field effects by computing the absolute signal propagation path length to each element of the antenna array. Because signals are difficult to penetrate buildings, we disable transmission through objects in outdoor scenarios and limit the number of transmissions to 1 in indoor scenarios.

Simulation data processing. Given the large number of channel samples to generate, simulating a large antenna array with many elements is time-consuming, even with ray tracing. Therefore, we only simulate the channel between a single pair of transmitter and receiver antennas and then extrapolate it to the channel of the whole antenna array. We assume that all antennas in the array experience the same propagation paths. Therefore, the channel response of an arbitrary antenna can be derived from its relative position to the reference antenna and the directions of the propagation paths. To address the near-field discrepancies when the array is too close to the sources, we trace back along the rays and identify the locations of the sources, *e.g.*, transmitter antenna, diffraction point, or transmission point, and then calculate the absolute distance between the source and each antenna in the array. To further address the polarization mismatch caused by different antenna array orientations, we simulate both vertically and horizontally polarized antennas.

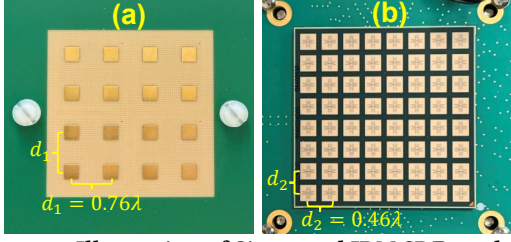


Figure 8: Illustration of Sivers and IBM SDR testbeds

Then, the channel for any array orientation can be derived by interpolating between the two basic polarization cases, thanks to the linearity of the wireless channel.

The final dataset consists of CSI and beam RSS samples of 16×16 uniform rectangular arrays (URA) with 0.5λ antenna spacing. The orientation of the antenna array is constrained to ensure at least one prominent signal path remains within the field-of-view of the antenna array, *i.e.*, elevation $\theta \leq 75^\circ$. A CSI sample is a 256×128 tensor, storing the channel response of 256 antennas and 128 sub-carriers. Beam RSS measurements are generated by applying the beam weight vectors to the CSI sample.

5.2 Training Procedure

BEAMFORMER is trained on NVIDIA RTX A100 GPUs with PyTorch. The Hugging Face Accelerate library for multi-GPU parallel training is used to efficiently process large-scale datasets comprising about 600k CSI samples per scenario.

Staged training. We employ a three-stage training scheme for all modules of BEAMFORMER. First, we pre-train the Beam Pattern Encoder and Latent Beam Processor for 3 epochs using a large number of data samples, with reference beams generated using random beam weight vectors. This stage establishes the fundamental feature extraction capability of BEAMFORMER. Second, we jointly train the Beam Pattern Encoder and Latent Beam Processor with the Beam Generator for 30 epochs. This stage refines the feature extraction capability and optimizes the reference beams of BEAMFORMER. Finally, the Beam Power Estimator is trained for 10 epochs to predict the scaling factor $\hat{\alpha}$ for denormalizing beam RSS. The MSE loss functions that minimize the RSS prediction errors are used in all three stages.

Training configurations. Cosine annealing learning rate scheduling with warm-up phases is used for training. Specifically, the warm-up duration is set to 5% of the training steps. The initial learning rate is set to 10^{-5} for the Beam Pattern Encoder and Latent Beam Processor, 10^{-4} for the Beam Generator, and 10^{-5} for the Beam Power Estimator. We use the AdamW optimizer with a weight decay coefficient of 0.01, and apply L2 regularization to enhance model generalization. This training configuration balances model convergence and computational efficiency.

6 Experiment

6.1 Experimental Setup

We evaluate BEAMFORMER with both simulated and real data. The simulation dataset generated at various frequencies and locations (Sec. 5) is split into training (600k samples) and testing (1k samples) sets. Since FR3 testbeds with large phased arrays are not commercially available, we collect the real dataset using two 28 GHz SDR



Figure 9: Experiment Outdoor and Indoor scenarios.

testbeds, Sivers EVK02004 [48] and IBM phased array antenna module (PAAM) [19], for evaluation only. Fig. 8 shows the phased arrays of the two SDRs. The Sivers SDR features a 4×4 phased array with an antenna spacing of 0.76λ . The real and imaginary components of each antenna weight have a $1/15$ quantization precision, enabling accurate phase control. The baseband signal is generated and processed by the USRP B210 SDR on both the transmitter and receiver sides. A PC controls both Sivers and USRP SDRs to synchronize beam switching and signal transmission. On the other hand, the IBM PAAM SDR features an 8×8 phased array with an antenna spacing of 0.46λ and 6-bit phase resolution for fine-grained beam pattern generation. Baseband processing is implemented with a USRP X310, which is synchronized with the PAAM using a 10 MHz reference clock and a PPS signal drawn from the USRP X310. Thanks to the tight synchronization, IBM PAAM SDR scans beams with a switching period of only $10\mu\text{s}$. The user equipment device is set to quasi-omnidirectional mode with PlutoSDR to mimic a small handheld form factor. Fig. 9 shows examples of outdoor and indoor scenarios in which data samples are collected.

We use the following primary metrics to evaluate the beam management performance. Specifically, the *RSS loss* between the globally optimal beam from brute-force search and the beam selected by the model is used to evaluate overall performance. In addition, the *RSS error* between the ground-truth and predicted RSS and the *AoD error* of dominant paths are used to evaluate the accuracy of the model. Note that RSS is an important indicator of the link data rate across different protocols.

6.2 Overall Performance

We compare the performance of BEAMFORMER with multiple baseline approaches across multiple simulation datasets of a 16×16 URA. (i) *Beam sweeping* measures the RSS of 64 predefined beams and selects the one with the highest RSS. Three variants are considered: Coarse beam sweeping uses only a 4×4 sub-array, similar to Sector Level Sweep in WiGig. Fine beam sweeping uses all 16×16 antennas. Hierarchical sweeping executes two rounds of beam sweeping, using 32 coarse beams during the first round and 32 fine beams within the main lobe direction of the optimal coarse beam during the second round, similar to Sector Level Sweep and Beam Refinement Phase in WiGig. (ii) *2ACE* [49] follows the compressive sensing principle to recover the CSI from beam measurements and subsequently derives the optimal beamforming weight vector. Due to the complexity of complete CSI reconstruction, we use reference 512 beams, aligning with the optimal configuration of 2ACE. (iii) *AgileLink* [20] is a beam searching scheme that uses multi-armed beams to achieve logarithmic complexity. The scheme requires $C \times K \times \log N$ beam measurements, where K is the num-

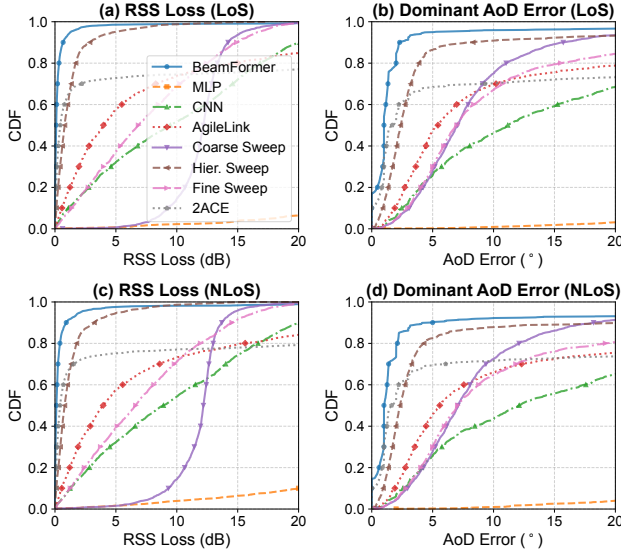


Figure 10: Overall performance of different approaches

ber of propagation paths, N is the number of antennas, and C is a constant coefficient. We set $C=6$, $K=8$, and $N=256$, leading to 384 beam measurements for AgileLink. (iv) CNN is based on the UNet [44] model, with a comparable size to BEAMFORMER. The input is a sampled beam spectrum, with only RSS measurements of the reference beams filled. The output is the reconstructed beam spectrum. (v) MLP replaces BEAMFORMER’s Latent Beam Processor with an MLP-based autoencoder [22], with the remaining parts and total parameter count consistent with BEAMFORMER.

Fig. 10 shows that BEAMFORMER outperforms all baseline approaches under both LoS and NLoS conditions with lower RSS loss and AoD error. Specifically, 90% of the prediction results of BEAMFORMER achieve RSS loss below 0.66 dB and AoD error below 2.29° . In NLoS scenarios, BEAMFORMER only experiences a slight performance degradation despite more complex multipath conditions, with the 90th percentile RSS loss and AoD error increasing to 0.92 dB and 4.54° , respectively. A common case of large AoD estimation errors occurs when multiple propagation paths have similar signal strengths, especially in indoor environments with strong multipath effects. Nonetheless, the RSS loss remains low in these cases.

In contrast, the baseline approaches exhibit various limitations. Coarse beam sweeping suffers from poor angular resolution, while fine beam sweeping has limited coverage. Hierarchical beam sweeping avoids these limitations by combining the two schemes and achieves performance comparable to BEAMFORMER. However, it sometimes only reaches local optima due to imperfect coarse beam sweeping, resulting in lower performance than BEAMFORMER. In addition, multiple rounds of beam sweeping incur significant latencies. 2ACE relies on the complete CSI reconstruction, which may have large errors due to limited beam measurements and the large number of antennas. AgileLink discretizes beam directions according to the number of antennas, and the quantization errors are exacerbated when both azimuth and elevation angles are considered with the URA. MLP, characterized by its simplistic fully-connected structure, struggles to capture the complex, high-dimensional mapping inherent in wireless channels. Finally, CNN captures only local

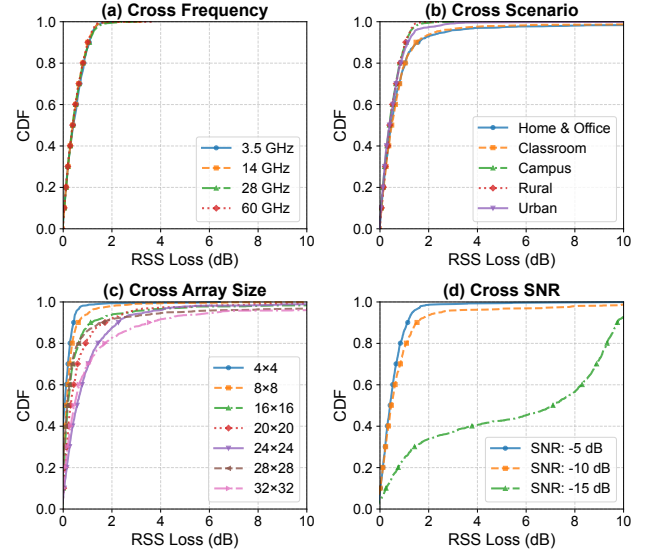


Figure 11: Impact of scene configurations

spatial correlations while remaining agnostic to beam weight information, limiting its ability to model global beam interactions and complicated beam pattern characteristics.

6.3 Performance Analysis

Next, we analyze the performance of BEAMFORMER across different configurations, including carrier frequencies, scenarios, antenna radiation patterns, and noise levels.

Impact of carrier frequencies. Fig. 11a illustrates the cross-frequency generalization capability of BEAMFORMER trained on the 28 GHz dataset and evaluated on datasets synthesized for different carrier frequencies, *i.e.*, 3.5 GHz, 14 GHz, 28 GHz, and 60 GHz. The results show that BEAMFORMER attains consistent performance across these bands without retraining. It indicates that BEAMFORMER captures the invariant structural characteristics of the wireless channel across different frequency bands.

Impact of scenarios. Fig. 11b shows the cross-scenario generalization capability of BEAMFORMER trained on an indoor dataset and evaluated in various scenarios, including indoor home, office, and classroom, and outdoor campus, rural, and urban regions. BEAMFORMER experiences slightly higher RSS loss in indoor scenarios, which typically have severer multipath effects. In contrast, while trained with the indoor dataset, BEAMFORMER performs better in outdoor scenarios with fewer propagation paths. Overall, BEAMFORMER achieves consistently high performance across scenarios.

Impact of antenna array settings. We evaluate the performance of BEAMFORMER across different antenna array sizes. For each antenna array size, we proportionally scale the number of model parameters, input beam pattern resolution, number of reference beams, and output beam spectrum resolution. As shown in Fig. 11c, BEAMFORMER achieves consistent performance, with only a slight increase in RSS loss as the array size grows. It is mainly due to the fact that larger arrays require substantially more model parameters and longer training time, resulting in insufficient convergence of the models due to our limited available computational resources.

As the training dataset of BEAMFORMER is generated using an

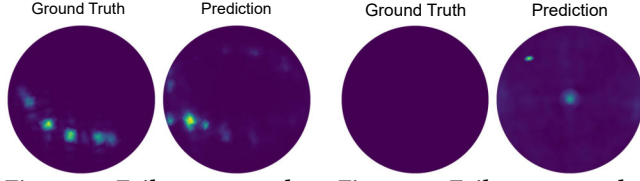


Figure 12: Failure case under severe multi-paths

Figure 13: Failure case under pure noise

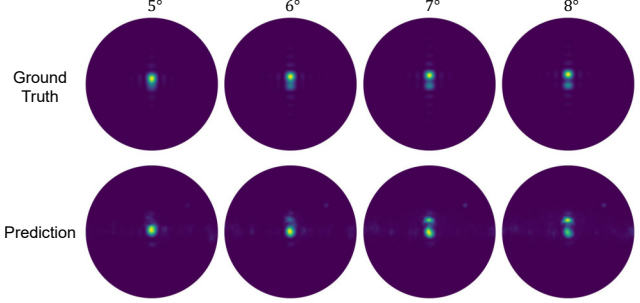


Figure 14: Beam Spectrum Resolution

array of isotropic antennas to reduce data synthesis overhead, we further evaluate the performance of BEAMFORMER with real patch antenna models. The results show that BEAMFORMER achieves comparable performance across both antenna types, indicating that antenna radiation patterns have a limited statistical impact on wireless channel characteristics.

Impact of noise levels. Fig. 11d shows the performance of BEAMFORMER at different noise levels. To simulate receiver noise, we add complex Gaussian noise to the CSI samples, with noise power determined by the average CSI power and the target SNR. Note that the beamforming gain is not included, as it depends on the specific beamforming configuration and varies significantly. The results show that the model retains low RSS loss at SNR levels of -5 dB and -10 dB. In contrast, the SNR of -15 dB is too weak to obtain clear RSS measurements, resulting in a dramatic performance drop. The results validate that BEAMFORMER operates robustly under typical SNR regimes of wireless communication systems [49].

Edge Cases. While BEAMFORMER is extensively trained to ensure stable convergence, it remains susceptible to extreme conditions, compared with the deterministic beam sweeping approaches. Specifically, severe multipath effect may lead to an erroneous beam spectrum, as shown in Fig. 12. Possible solutions to address this type of failure include increasing model capacity and balancing the light and severe multipath effect cases in the training dataset. Besides, noise-only input with extremely low SNR levels may trigger the generation inconsistency of transformers, which yield phantom paths, as shown in Fig. 13. Fortunately, such failures can be avoided by monitoring the input RSS levels of the reference beams.

Since BEAMFORMER is trained with the synthetic data from a 16×16 antenna array, its angular resolution should be bounded by the antenna array. To evaluate it, we simulate cases with two signal paths that are closely spaced in the angular domain. As illustrated in Fig. 14, BEAMFORMER can separate the two paths until their angular spacing reaches 6° , below which the two paths are hallucinated into a single merged path. The resolution limit of BEAMFORMER aligns with that of a 16×16 URA, *i.e.*, 6.4° .

Multipath prediction accuracy. Fig. 15 shows the RSS error across

Scheme	Measurement Latency		Inference Latency
	Infinite SSB burst	Finite SSB burst	
BeamFormer	1.1	1.1	1.8
MLP	1.1	1.1	1.3
CNN	1.1	1.1	5.9
AgileLink	6.9	101.1	4.3
Beam Sweeping	1.1	1.1	0
2ACE	9.1	141.1	4248

Table 2: Latency (ms) of different approaches

the prediction of the first three dominant propagation paths. BEAMFORMER achieves consistently low RSS error for the dominant path, demonstrating its effectiveness for optimal beam selection. However, the prediction accuracy gradually drops for the secondary and tertiary paths, as they have less impact on the input reference beam measurements and are more sensitive to output errors.

6.4 Model Analysis

This section evaluates the contribution of each key design component of BEAMFORMER.

Positional encoding. We compare the beam pattern encoding scheme with two baselines. The first baseline directly encodes complex-valued beam weights, while the second only encodes main lobe directions with the standard 2D sinusoidal encoding. Fig. 18 shows that using only the main lobe direction fails with significant RSS errors, as it disregards critical beam pattern information that impacts RSS. In contrast, beam weight encoding achieves performance similar to that of beam pattern encoding. However, it omits the antenna array configurations and cannot be generalized to different antenna array geometries.

Processing latencies. We then evaluate the benefits of latent beam processing, compared with the standard transformer. Fig. 16 shows the processing latencies of the two schemes. As the number of query beams increases and the beam spectrum resolution becomes finer, the latency of the standard transformer increases quadratically. In contrast, the latent beam processing retains low latencies, thanks to its linear complexity. The latency of BEAMFORMER can be further reduced through parallel query processing on GPUs and caching of beam pattern encodings. With an inference time of 1.8 ms, BEAMFORMER potentially meets the latency requirements of beam management in wireless networks.

Tab. 2 compares the measurement and inference latencies of different approaches. We consider the measurement latencies with both an ideally infinite number and a practically finite number (*i.e.*, 64) of SSBs within the typical 20 ms periodicity of the SSB burst in 5G. The duration of each SSB is $17.84 \mu\text{s}$ when the subcarrier spacing (SCS) is 240 kHz [16]. Therefore, the ideal latency is $0.01784N$ ms while the practical latency is $20 \lceil \frac{N-1}{64} \rceil + 0.01784(\lceil \frac{N-1}{64} \rceil + 1)$ ms, where N is the number of reference beams. Compared with beam sweeping baselines, BEAMFORMER introduces only an additional 1.8 ms inference latency while eliminating the need for brute-force reference beam measurements to achieve full angular coverage. Moreover, the inference latency of BEAMFORMER is comparable to that of MLP and much lower than that of a CNN. In contrast, both AgileLink and 2ACE have higher beam measurement latencies, and 2ACE has significantly higher inference latency due to time-

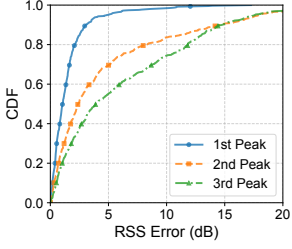


Figure 15: Multi Path Prediction Accuracy

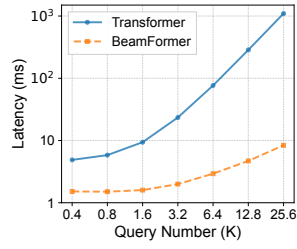


Figure 16: Comparison of model latencies

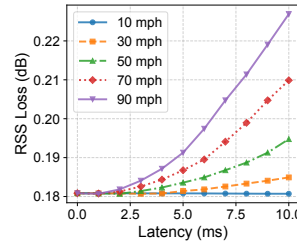


Figure 17: Impact of latency and user mobility

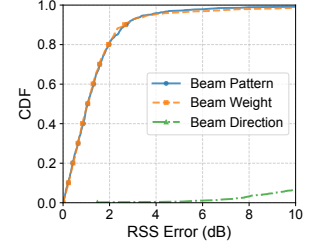


Figure 18: Comparison of positional encoders

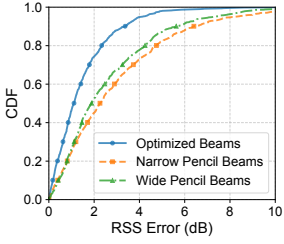


Figure 19: Comparison of reference beam settings

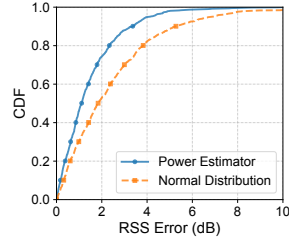


Figure 20: Comparison of power estimators

consuming optimization.

Given the latency range of BEAMFORMER, we further evaluate the RSS loss of beam misalignment caused by the movement of user equipment. Specifically, we move the user equipment in simulation scenarios based on the latency and user speed and resimulate the scenarios to obtain RSS losses. As shown in Fig. 17, even with a maximum latency of 10 ms (*i.e.*, 25,600 query beams) and a maximum speed of 90 mph, the RSS loss is only around 0.23 dB, demonstrating the robustness of BEAMFORMER under high-mobility conditions.

Reference beam configurations. To evaluate the contribution of the universal reference beam optimization, we compare the optimized beams with conventional pencil beams, including narrow beams using the entire antenna array and wide beams using a 4x4 sub-array. As shown in Fig. 19, the reference beams optimized by the Beam Generator achieve the lowest RSS error, as they optimally steer transmit power according to the latent patterns of multipath channels. In contrast, the narrow and wide pencil beams exhibit higher RSS errors. On the one hand, due to the high angular resolution of narrow beams, BEAMFORMER generates more accurate normalized beam spectra, but struggles to recover the absolute peak power. On the other hand, due to the broader angular coverage of wide beams, BEAMFORMER achieves more accurate power estimation, but the beam spectra generated have limited resolution.

Absolute power estimation. We compare the Beam Power Estimator with a statistical baseline that draws power ratios $\tilde{\alpha}$ from a normal distribution fit to the training dataset. As shown in Fig. 20, the Beam Power Estimator achieves more accurate power prediction, as it successfully maps the RSS variations of reference beams to absolute power ratios for denormalizing beam spectra.

Model parameters. We conduct comprehensive evaluations regarding key model parameters, including the number of reference beams, feature dimension, number of processing layers, and number of latent tokens. First, Fig. 21a shows that RSS error slightly

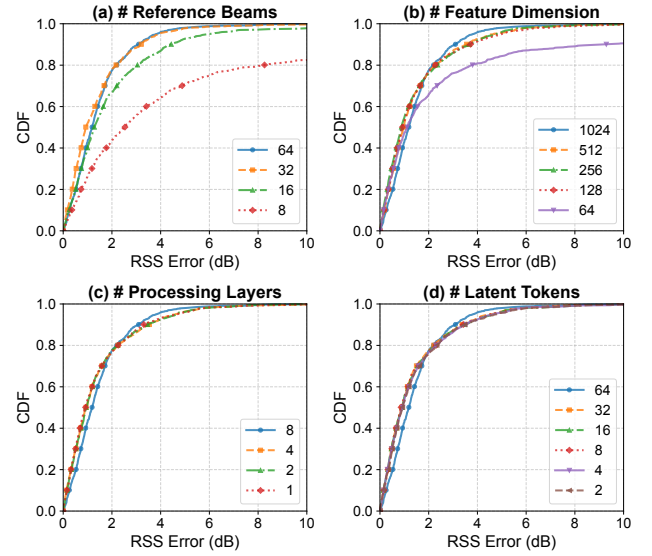


Figure 21: Impact of model parameters

increases as the number of beams reduces from 64 to 16. The error becomes pronounced when the beam number is reduced to 8, as fewer reference beams provide less channel information. Second, the error increases significantly only when the feature dimension is reduced to 64, where the model capacity becomes insufficient to capture beam-wise characteristics. Third, the model performance does not change significantly with the other two parameters, indicating that a lightweight transformer model with fewer processing layers and latent tokens is sufficient for beam spectrum generation.

6.5 Case Study

To evaluate the performance of BEAMFORMER in real-world environments, we collect data in outdoor LoS and indoor LoS/NLoS scenarios using the two 28 GHz SDRs (See Fig. 9). Outdoor NLoS scenarios are not considered, as the indoor NLoS cases already capture more challenging multipath conditions. For the Sivers SDR, we collect 216 beam measurements at each location, including 16 reference beams generated by the Beam Generator and 200 uniformly sampled pencil beams for ground-truth beam spectrum. For the IBM PAAM SDR, RSS measurements of 32 reference beams and 416 pencil beams are collected. During data collection, we randomly rotate the receiver while ensuring that at least one dominant path falls within its field of view. We notice that the transmit power of both SDR testbeds varies with the number of activated antennas and

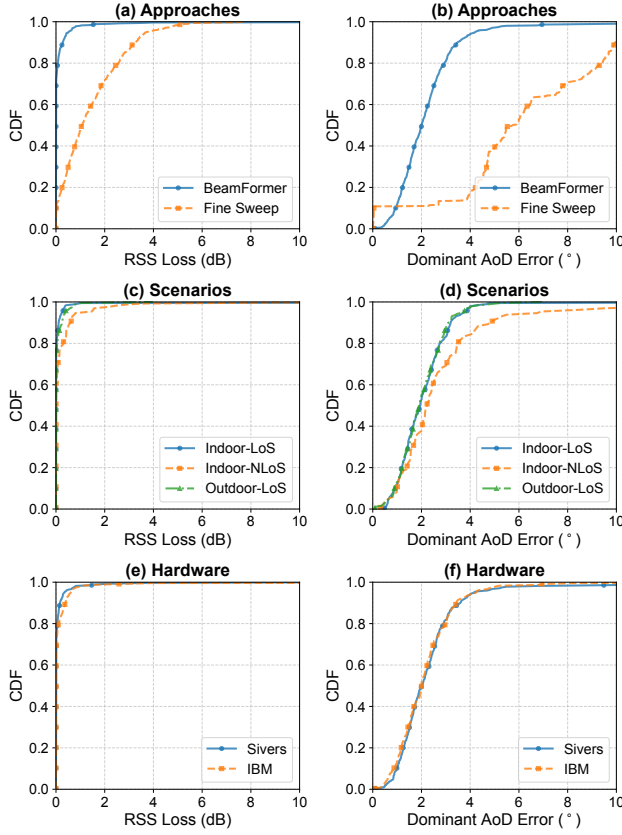


Figure 22: Performance with real-world data

is difficult to calibrate. Therefore, we keep all antennas activated in reference beam configurations to mitigate the transmit power variations. On the other hand, phase offsets between antenna elements are calibrated and incorporated into the actual beam weight vectors. Fine beam sweeping with the same number of reference beam measurements is used as the baseline for comparison. The path distributions of different scenarios are summarized in Tab. 3.

Fig. 22a and b show the performance of BEAMFORMER compared to the fine beam sweeping scheme, using both SDRs. BEAMFORMER achieves accurate beam spectrum reconstruction, with RSS loss lower than 1 dB and AoD errors below 4° for 95% test samples. In contrast, the fine beam sweeping scheme experiences greater RSS loss with a longer tail beyond 5 dB, and the AoD errors are mainly around 4° – 10° . Note that Sivers SDR only has 16 antennas, and IBM PAAM SDR only has 64 antennas. The actual performance of the fine beam sweeping scheme with larger antenna arrays is expected to further deteriorate, as indicated in Fig. 10.

Fig. 22c and d show the performance of BEAMFORMER in three scenarios, outdoor LoS, indoor LoS, and indoor NLoS conditions. BEAMFORMER achieves low RSS loss and AoD errors under LoS conditions. In contrast, the errors become slightly higher in NLoS scenarios. The performance degradation is primarily due to the increasing number of dominant propagation paths and lower SNR. As the underlying channel structure becomes more complex, reconstructing the beam spectrum with BEAMFORMER becomes increasingly challenging. Fortunately, FR2/FR3 channels, where beam

Device	Scenario	Total	Path Number		
			1	2	≥ 3
Sivers	Indoor-LoS	114	99.1%	0.9%	0.0%
	Indoor-NLoS	56	53.6%	26.8%	19.6%
	Outdoor-LoS	108	100.0%	0.0%	0.0%
IBM	Indoor-LoS	77	80.5%	19.5%	0.0%
	Indoor-NLoS	76	64.5%	28.9%	6.6%
	Outdoor-LoS	83	96.4%	3.6%	0.0%

Table 3: Path distribution across scenarios

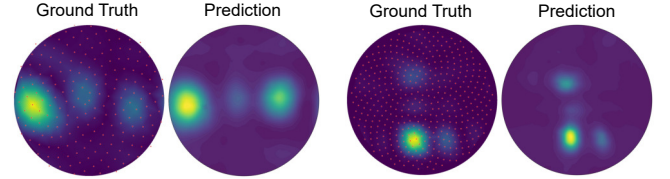


Figure 23: Example spectrum with Sivers SDR

Figure 24: Example spectrum with IBM SDR

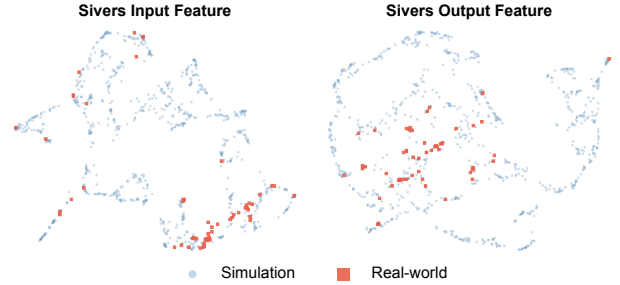


Figure 25: UMAP Visualization of Simulation and Real-World Feature Distributions

management is needed, are likely to be sparse with a few dominant propagation paths only [20].

We further compare the performance of BEAMFORMER with the two SDRs. As shown in Fig. 22e and f, despite different antenna array configurations, BEAMFORMER achieves consistently low RSS loss and AoD errors. Fig. 23 and 24 visualize two examples of beam spectra collected by Sivers and IBM PAAM SDRs, respectively. BEAMFORMER successfully reconstructs the dominant propagation paths. The results indicate that BEAMFORMER can generalize across heterogeneous phased array hardware.

The comparable performance of BEAMFORMER on both simulated and real data indicates strong statistical similarity between the two datasets. To validate it, we apply UMAP [34] to visualize the feature distributions of BEAMFORMER encoder inputs and decoder outputs. As shown in Fig. 25, the simulation and real data form overlapping clusters in both the input and output feature spaces, indicating a minor sim-to-real gap. Therefore, BEAMFORMER can be trained solely on simulation data while remaining generalizable to real-world deployments without the need for domain adaptation.

7 Discussion

Universal reference beam optimization. BEAMFORMER uses a set of universal reference beams optimized on a large training dataset. However, these beams may be suboptimal in specific scenarios and under channel dynamics. One potential augmentation of BEAMFORMER is a reference beam selection policy based on beam

measurements and BEAMFORMER outputs from previous beam management cycles. Deep reinforcement learning can be adopted to achieve accurate and fast beam adaptation, thereby enhancing the robustness of BEAMFORMER in dynamic or unseen scenarios.

Sim-to-real gap. BEAMFORMER is trained on simulation datasets, which often exhibit domain shifts relative to real-world data. By carefully crafting simulation scenarios, the statistical sim-to-real gap is largely mitigated. As a result, significant beam spectrum generation errors occur only under complex multipath conditions, which are rare even in indoor scenarios at FR2/FR3 frequencies. A straightforward solution is to fine-tune BEAMFORMER with a small real-world dataset, whose collection overhead is reasonable and can be amortized across the long-term operation of base stations.

Application of interference management. With the reclaim of the FR3 band, 6G faces the coexistence issues with incumbent services. Using conventional beam sweeping schemes, 6G networks may miss viable spectrum sharing opportunities due to limited beam measurements. In contrast, the generative capability of BEAMFORMER potentially benefits interference management of 6G. Specifically, BEAMFORMER can use reference beams with low gain in the directions of incumbent devices to limit the interference during the beam searching process. Then, using the reconstructed beam spectrum, the beam with the highest gain and satisfying the interference constraints can be selected.

8 Related Work

Beam management. Beam management in wireless networks seeks to determine the optimal beam with the highest RSS from a set of reference beams. Hierarchical search strategies [9, 36, 39, 55] have been developed to achieve logarithmic numbers of beam measurements. Intuitively, hierarchical search performs multiple rounds of beam measurements, where each round focuses on the optimal direction identified in the previous round. However, hierarchical search usually uses low-gain wide beams in early rounds, which are vulnerable to noise and may lead to suboptimal search spaces. Besides, the processing and feedback across multiple rounds introduce significant latency. Instead of using common pencil beams, existing works [2, 20, 54] have explored nontrivial beam patterns to reduce overhead. AgileLink [20] uses multi-armed beams with random beam weight permutation strategies to achieve efficient sampling in the angular domain. However, it restricts beamforming directions to a discrete angular grid, leading to misalignment in the continuous angular space and thereby degrading performance. Some works utilize out-of-band modalities [3, 12, 33, 58] at the cost of hardware complexity. In addition, the out-of-band channels have nonnegligible discrepancies from the in-band channel. For example, compared to FR2 and FR3 frequencies, sub-6 GHz [3] has stronger diffraction, different material reflectance and transmittance, and lower angular resolution due to smaller antenna arrays. Lidar [58] faces limited coverage due to visual blockages, making it unreliable in obstructed environments. Radar [12] faces the challenge of identifying actual user equipment from the sensing targets.

Beamforming with CSI. Fine-grained CSI can assist beam management by revealing the globally optimal beamforming configurations. For example, Orthogonal Matching Pursuit (OMP) [29, 56] works by sequentially identifying the strongest propagation path

from the residue channel response. Thresholding algorithms, *e.g.*, Iterative Hard Thresholding (IHT), apply hard thresholds to CSI estimation during iterative updates to approximate sparse channels [7]. Statistical approaches, *e.g.*, Bayesian frameworks, use prior distributions [40, 45, 47] to model channel sparsity. However, these approaches require coherent CSI measurements, which are typically unavailable with phased arrays. Some recent works [38, 49] reconstruct CSI from incoherent RSS measurements. 2ACE [49] identifies dominant propagation paths through RSS measurements to reconstruct CSI. ACO [38] extracts the CSI per antenna from the beams whose corresponding antenna weight varies only. Despite accurate CSI estimation, these approaches require a large number of beam measurements, leading to unacceptable latencies in practice. State space based approaches, *e.g.*, Kalman filter [46], have been used for beam refinement to address UE mobility and environmental variations. They are complementary to BEAMFORMER as the latter focuses on initial beam acquisition.

AI-based channel reconstruction. Recent studies employ AI techniques to capture the super-resolution geometry of the environment and reconstruct the wireless channel. For example, some works [4, 5, 32] use neural networks to refine distance estimates for the dominant propagation paths to assist cross-frequency CSI estimation. Yet applying these methods to phased arrays remains underexplored. On the other hand, neural representations have been adopted for wireless environment reconstruction and channel modeling [10, 11, 37, 57, 59]. For example, NeRF2 [59] extends neural radiance fields to learn the continuous volumetric RF field for a physical environment. GSRF [57] develops complex-valued 3D Gaussian Splatting to model directional and phase-dependent radiance. Despite their high reconstruction accuracy, the instances of these models are tied to specific scenarios in which their training data were collected and cannot be easily adapted. BEAMFORMER differs fundamentally from these neural representations in that it captures the general relations among beam measurements and is not restricted to any specific environment.

9 Conclusion

Through BEAMFORMER, we have verified the feasibility of using generative transformer models for beam spectrum generation. Conventional transformers cannot encode beam information, suffer from high latencies, and are prone to generation errors. BEAMFORMER addresses these challenges through the designs of beam pattern encoding, latent beam processing, and universal reference beam optimization. BEAMFORMER can be seamlessly integrated into the beam management process to overcome the prohibitive overhead associated with denser antenna arrays in future 6G.

10 Acknowledgement

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A Artifact Appendix

A.1 Abstract

This artifact contains the source code, pre-trained model weights, and datasets to reproduce Fig. 10-25. The models consist of the four jointly trained components, i.e., Beam Pattern Encoder, Latent Beam Processor, Beam Generator, and Beam Power Estimator, with the pre-trained weights for all experimental configurations. The datasets comprise simulated CSI generated via Wireless InSite ray-tracing across multiple frequency bands (3.5/14/28/60 GHz), antenna array configurations (4×4 to 32×32 URA), and indoor/outdoor scenarios, as well as real-world 28 GHz beam-wise CSI collected from Sivers EVK02004 and IBM PAAM SDR testbeds. All software dependencies are packaged in a Docker image, and evaluation is fully automated via Python and Bash scripts. The evaluation requires an NVIDIA GPU with ≥20 GB VRAM and reproduces performance comparison between BEAMFORMER and the baselines, design ablations, and generalization to real SDR hardware.

A.2 Artifact Check-List (Meta-Information)

- **Algorithm:** Beam spectrum generation with transformers; universal reference beam optimization
- **Program:** Python; PyTorch; Bash; MATLAB
- **Data set:** Included.
- **Run-time environment:** Docker; Linux; CUDA
- **Hardware:** Windows or Linux supporting NVIDIA GPU with ≥20 GB VRAM, e.g., RTX 4000 Ada and A40.
- **Execution:** `bash run_docker.sh`; fully automated; ≈2–3 hours
- **Metrics:** RSS Loss (dB); AoD Error (degrees); RSS Error (dB); inference latency (ms)
- **Output:** PDF/PNG (Fig. 10-25)
- **Experiments:** See Sec. A.4 for details.

- **How much disk space required (approximately)?:** ≈56 GB (datasets + checkpoints) + ≈20 GB (Docker image)
- **How much time is needed to prepare workflow (approximately)?:** ≈20 min (Docker build) + download time (≈56 GB)
- **How much time is needed to complete experiments (approximately)?:** ≈2–3 hours (from scratch); >12 hours for 2ACE baseline from scratch
- **Publicly available?:** Yes.
- **Code licenses (if publicly available)?:** MIT License.
- **Data licenses (if publicly available)?:** MIT License.
- **Workflow automation framework used?:** Yes, Bash + Python.
- **Archived (provide DOI)?:** Yes² (artifact evaluation submission).

A.3 Description

A.3.1 How to access. The latest artifact further covering the camera-ready changes is publicly available at GitHub³.

A.3.2 Hardware dependencies. An NVIDIA GPU with ≥20 GB VRAM and Docker with `--gpus all` support (NVIDIA Container Toolkit). The artifact has been verified on NVIDIA RTX 4000 Ada (20 GB, driver 573.44, CUDA 12.8) and A40 (46 GB, driver 580.95.05, CUDA 13.0). Approximately 76 GB of free disk space is required.

A.3.3 Software dependencies. All dependencies are encapsulated in the Docker image (base: `huggingface/accelerate:gpu-nightly`). Key packages installed inside the container: PyTorch, Accelerate, Scipy, Numpy, Matplotlib, Pandas. Reproducing the 2ACE baseline from scratch additionally requires MATLAB with the Python engine, where the details are clarified in `requirements.txt`.

A.3.4 Data sets. All datasets are included in `ae_necessary.zip` and can be downloaded by executing `prepare_files.sh`. The archive contains simulated CSI data across multiple frequency bands, antenna configurations, and indoor/outdoor scenarios generated via ray-tracing, as well as real-world 28 GHz CSI collected from Sivers EVK02004 and IBM PAAM. The full training dataset is publicly available on IEEE DataPort⁴, with the data processing code on GitHub⁵.

A.3.5 Models. Pre-trained weights for all model components, i.e., Beam Pattern Encoder, Latent Beam Processor, Beam Generator, and Beam Power Estimator, are included in `ae_necessary.zip` and cover all experimental configurations in the paper.

A.4 Installation

Step 1 – Build the Docker image.

```
docker build -t beamformer-docker .
```

Build the image before running `prepare_files.sh` to avoid copying large data files into the build context.

Step 2 – Download datasets and model checkpoints.

```
bash prepare_files.sh
```

Download `ae_necessary.zip` (≈56 GB) and extract and move the contents to the project root.

²<https://doi.org/10.5281/zenodo.19546148>

³<https://github.com/Shunqiang-Feng/BeamFormer/tree/AE>

⁴<https://doi.org/10.21227/g1zj-z323>

⁵<https://github.com/Shunqiang-Feng/BeamFormer-Dataset>

A.5 Experiment workflow

Run the full artifact inside Docker:

```
bash run_docker.sh
```

Reproduce Fig. 10-25 in a single run, with the results saved to `./ae_figures/`. An individual figure can be reproduced via:

```
python reproduce_figures.py --figure "<figure_name>" \
  --data_source from_scratch
```

Use `--list` to display all available figure names. `cache` (default) regenerates plots from pre-computed results; `from_scratch` re-runs model inference.

A.6 Evaluation and expected results

Upon successful completion, the figures are produced in `./ae_`

`figures/`, which should be consistent with the figures in the paper. It should be noted that the latency measurements in Fig. 16 may vary across machines, and the UMAP algorithm in Fig. 25 has inherent randomness. Nonetheless, the reproduced results should exhibit consistent patterns.

A.7 Notes

2ACE baseline in Fig. 10. 2ACE requires proprietary MATLAB software. Pre-computed results in `cache/2ACE/` are used by default. To reproduce from scratch: (1) install MATLAB with the Python engine; (2) delete `cache/2ACE/`; (3) re-run with `--data_source from_scratch`. Expected runtime is about 12 hours.